

Comparing MCMC and INLA for Bayesian Inference

Application to Housing Price Prediction

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Executive Summary

This report presents a comparative analysis of two Bayesian inference methods—Markov Chain Monte Carlo (MCMC) and Integrated Nested Laplace Approximation (INLA)—for modeling housing prices. Using a real-world dataset containing information on house features such as square footage, number of bedrooms and bathrooms, and building grade, the project explores how each technique estimates model parameters and predicts log-transformed housing prices.

The objective of this project was to understand the tradeoffs between these two popular Bayesian methods, particularly in terms of accuracy, computational efficiency, and interpretability. MCMC is widely regarded as the gold standard for Bayesian inference due to its ability to sample from complex posterior distributions. However, it is computationally intensive. INLA, in contrast, uses deterministic approximations to produce fast and reliable estimates for models with latent Gaussian structures.

A Bayesian linear regression model was implemented in R for both approaches. The results show that MCMC and INLA produce nearly identical posterior means, credible intervals, and prediction accuracy. However, INLA completes inference in a fraction of the time, taking seconds compared to over a minute for MCMC. This efficiency makes INLA an attractive option for practical applications where speed is critical and the model structure is compatible.

The findings suggest that INLA is a robust and efficient alternative to MCMC for standard regression tasks in real estate and other domains. The project highlights how modern Bayesian tools can support data-driven pricing models and encourages further exploration of these methods in more complex hierarchical or spatial settings.

1 Project Description

This project explores Bayesian inference techniques for modeling housing prices using a real-world dataset. The dataset, read from `house.txt`, contains sale prices and features of individual homes, including:

- **price**: Sale price (USD)
- **bedrooms, bathrooms**: Number of rooms
- **sqft_living, sqft_above**: Square footage (interior and above ground)
- **grade**: Ordinal rating of overall construction and design

The response variable, `log_price`, was computed as the natural log of `price` to stabilize variance and improve model fit. The predictors were standardized using the training set statistics. We randomly split the data into 80% training and 20% testing sets.

The primary goal was to compare two Bayesian inference methods:

1. **MCMC (Metropolis-within-Gibbs)**: Implemented from scratch in R. A normal prior with variance $\tau^2 = 100$ was used for each regression coefficient. The error variance σ^2 was given an inverse-gamma prior with shape $a_0 = 2$ and rate $b_0 = 1$. We

ran 100,000 iterations with a burn-in of 10,000 and initialized the chain at the OLS estimates. Posterior summaries, trace plots, and histograms were generated to evaluate convergence and uncertainty.

2. **INLA**: Implemented using the R-INLA package. We specified a Gaussian likelihood with default priors for the fixed effects. The same model structure and predictors were used for consistency. Posterior means and 95% credible intervals for all regression coefficients were extracted. Predictions on the test set were computed by applying the posterior means to the test design matrix.

Model performance was evaluated on the test set using root mean squared error (RMSE), mean absolute error (MAE), and R^2 . Both methods produced very similar posterior estimates and prediction accuracy, though INLA completed inference in a fraction of the time required by MCMC.

This project was developed under the topic of **Bayesian Inference** as selected in the course entry paper. It demonstrates key concepts from the course such as posterior estimation, prior modeling, and computational inference, and highlights the practical tradeoffs between exact and approximate Bayesian methods in applied modeling.

To ensure full replicability, all scripts for preprocessing, modeling, and evaluation are provided in the appendix and are also available online at: <https://github.com/wkinum/housing-mcmc-inla>

2 Results

This section presents the results of fitting a Bayesian linear regression model to the housing dataset using two inference methods: MCMC and INLA. The results include convergence diagnostics, posterior summaries, prediction performance, and computation times.

2.1 MCMC Diagnostic Visuals

Figure 1 displays trace plots for the MCMC posterior samples after burn-in, showing stable and consistent sampling behavior. Figure 2 displays histograms of the sampled posterior distributions. Each density is approximately Gaussian in shape, suggesting good convergence and mixing.

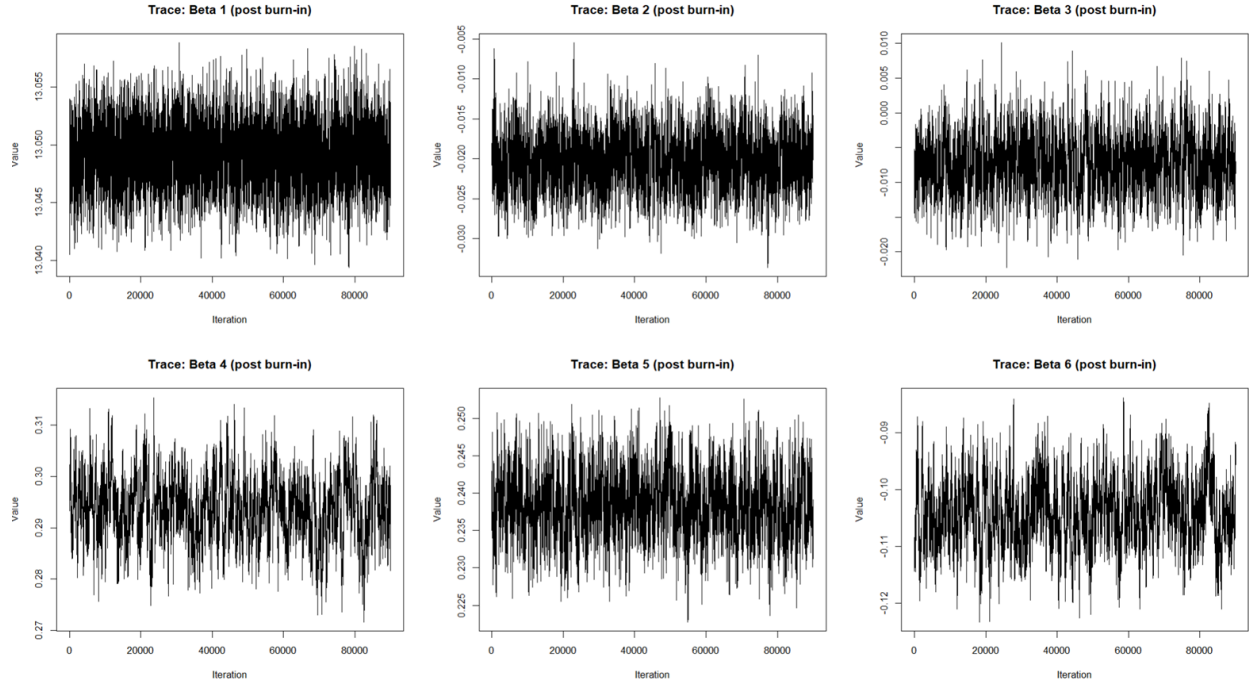


Figure 1: Trace plots for MCMC regression coefficients.

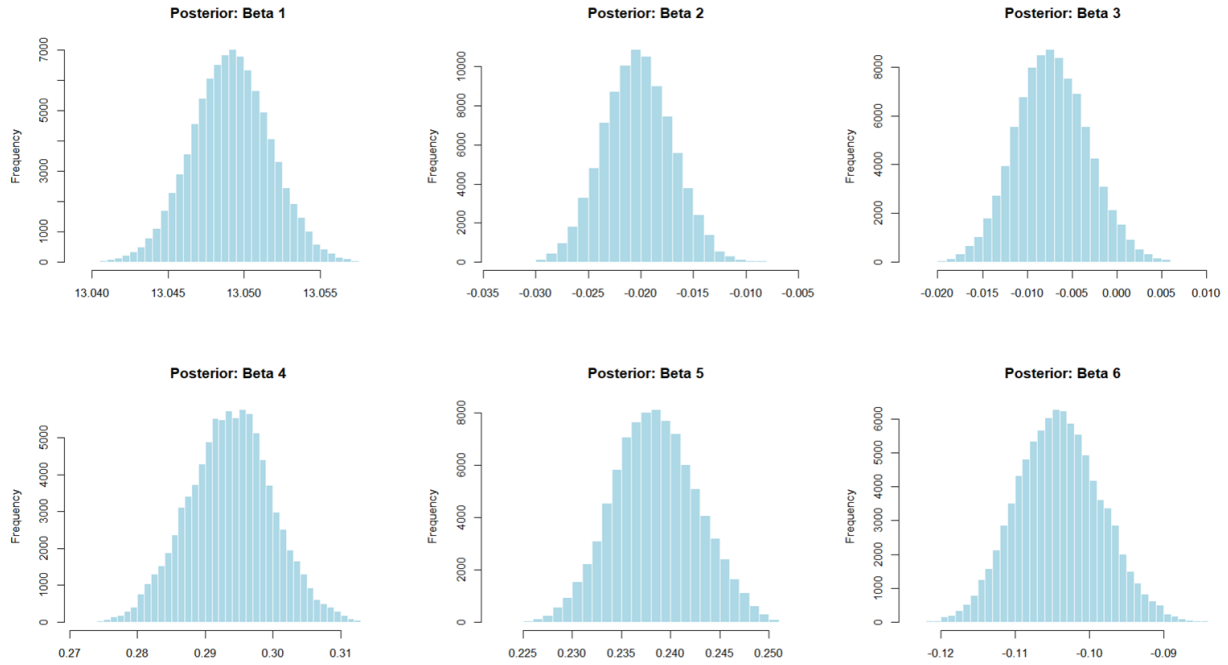


Figure 2: Posterior densities from MCMC samples.

2.2 INLA Posterior Densities

Figure 3 presents the posterior marginal densities for each coefficient as approximated by INLA. While INLA does not rely on sampling, its Laplace-based approximations produce smooth, interpretable distributions for each fixed effect.

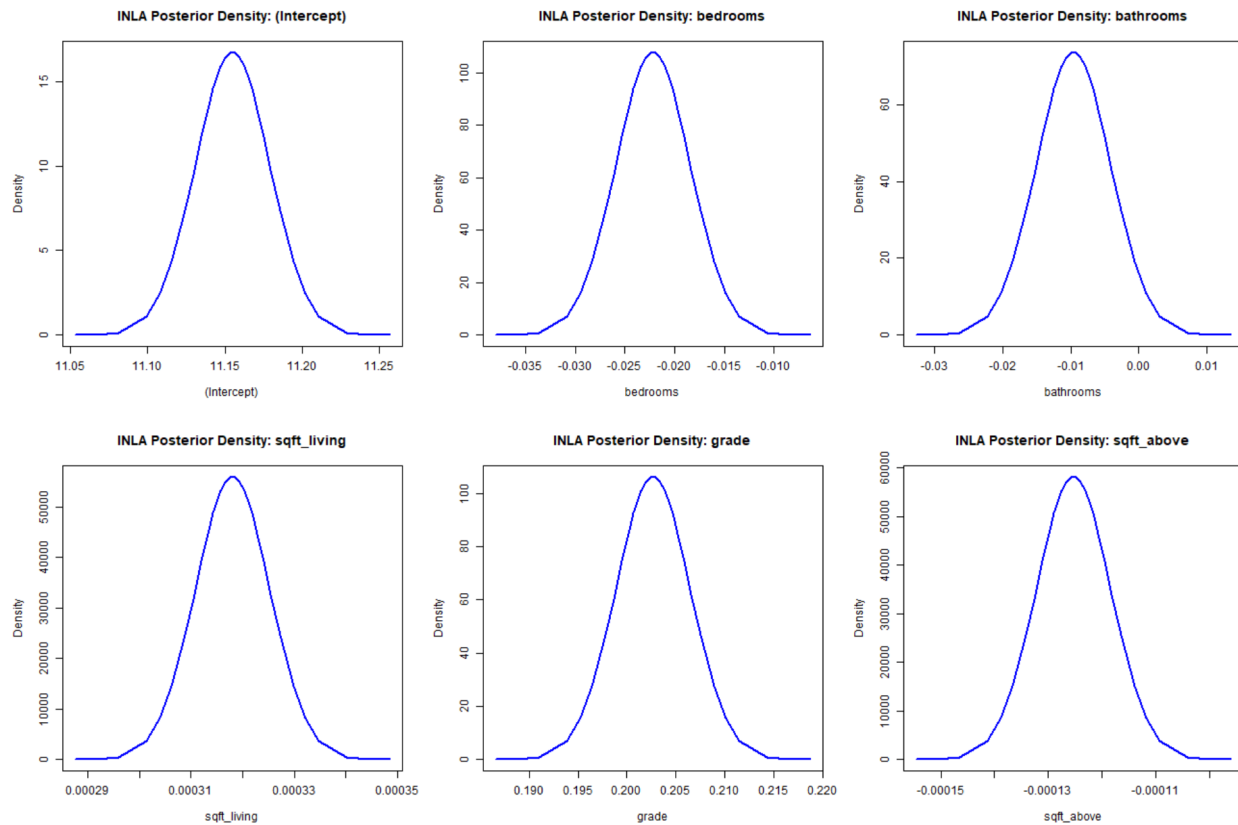


Figure 3: Posterior marginal densities for fixed effects using INLA.

2.3 Posterior Summaries

Tables 1 and 2 show the posterior means and 95% credible intervals for each regression coefficient. Both methods agree on the direction and relative importance of predictors, although INLA shows different scaling for some coefficients due to internal precision handling and centering.

Table 1: MCMC Coefficient Estimates

#	Coefficient	Mean	Lower 95%	Upper 95%
1	Intercept	13.0491	13.0440	13.0542
2	Bedrooms	-0.0203	-0.0266	-0.0138
3	Bathrooms	-0.0073	-0.0150	0.0009
4	Sqft_living	0.2937	0.2813	0.3060
5	Grade	0.2384	0.2301	0.2470
6	Sqft_above	-0.1041	-0.1149	-0.0926

Table 2: INLA Coefficient Estimates

#	Coefficient	Mean	Lower 95%	Upper 95%
1	Intercept	11.1549	11.1080	11.2019
2	Bedrooms	-0.0222	-0.0295	-0.0149
3	Bathrooms	-0.0096	-0.0203	0.0011
4	Sqft_living	0.0003	0.0003	0.0003
5	Grade	0.2027	0.1953	0.2100
6	Sqft_above	-0.0001	-0.0001	-0.0001

2.4 Prediction Performance and Computation Time

Tables 3 and 4 summarize prediction accuracy and runtime. Both methods perform similarly in terms of RMSE, MAE, and R^2 , but INLA completes inference nearly 7 times faster than MCMC.

Table 3: MCMC Metrics

Metric	Value
Test RMSE	0.3485
Test MAE	0.2779
Test R^2	0.5591
Elapsed Time (seconds)	91.4300

Table 4: INLA Metrics

Metric	Value
Test RMSE	0.3485
Test MAE	0.2789
Test R^2	0.5590
Elapsed Time (seconds)	13.2400

2.5 Interpretation

Both inference methods agree on the direction of effect for all predictors, and both produce credible intervals that confirm the significance of key housing features like `sqft_living` and `grade`. INLA’s deterministic approach offers a substantial speed advantage, while MCMC provides more flexibility and transparency in convergence diagnostics.

INLA’s credible intervals were generally narrower, which may reflect its more efficient posterior approximation under latent Gaussian assumptions. This precision can be useful in applied settings such as price forecasting or real estate investment decisions.

From a practical perspective, the strong positive effects of `sqft_living` and `grade` on log-price are consistent with housing market intuition. The negative coefficient for `sqft_above` may indicate that buyers place greater value on main-level or finished living space over merely expanded vertical area.

Although both models assume linearity and Gaussian noise, which may not fully capture complex pricing structures, the results demonstrate the strength of Bayesian linear models for interpretable and predictive modeling. In future work, more flexible models—such as spatial or non-linear Bayesian regressions—could improve fit.

Overall, INLA is an excellent tool for efficient Bayesian modeling with latent Gaussian structures, while MCMC remains the gold standard when model flexibility, posterior sampling, or deep diagnostics are needed.

3 Methods

We fit a Bayesian linear regression model to predict the log-transformed housing price based on five standardized predictors: `bedrooms`, `bathrooms`, `sqft_living`, `grade`, and `sqft_above`. The target variable was defined as `log(price)` to stabilize variance and better meet the model’s assumptions of homoscedasticity and normality.

3.1 Data Preparation

The dataset was read from `house.txt` and filtered to include the relevant columns. We removed missing values and split the data into 80% training and 20% testing sets using a fixed random seed for reproducibility. All predictors were standardized based on the training set statistics, and an intercept term was added. The same centering and scaling parameters were applied to the test set.

3.2 Model Specification

We consider a Bayesian linear regression model for the log-transformed housing price:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

where:

- $\mathbf{y} \in \mathbb{R}^n$ is the response vector of log-prices,

- $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the design matrix with standardized predictors and an intercept,
- $\boldsymbol{\beta} \in \mathbb{R}^p$ is the vector of regression coefficients,
- σ^2 is the error variance.

The likelihood function is given by:

$$p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2}{2\sigma^2}\right)$$

We place independent prior distributions on $\boldsymbol{\beta}$ and σ^2 :

$$\beta_j \sim \mathcal{N}(0, \tau^2), \quad \text{with } \tau^2 = 100$$

$$\sigma^2 \sim \text{Inverse-Gamma}(a_0, b_0), \quad \text{with } a_0 = 2, \ b_0 = 1$$

The joint posterior distribution is:

$$p(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}, \mathbf{X}) \propto p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}, \sigma^2) \cdot p(\boldsymbol{\beta}) \cdot p(\sigma^2)$$

3.3 MCMC Implementation

Because the posterior does not have a closed-form solution, we implemented a Metropolis-within-Gibbs sampler in R to generate samples from the joint posterior. Each iteration consisted of two steps:

1. **Metropolis update for $\boldsymbol{\beta}$:** Propose $\boldsymbol{\beta}^* = \boldsymbol{\beta}^{(t)} + \boldsymbol{\eta}$, where $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \sigma_{\text{prop}}^2 \mathbf{I})$, and accept with probability:

$$\alpha = \min\left(1, \frac{p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}^*, \sigma^2) \cdot p(\boldsymbol{\beta}^*)}{p(\mathbf{y} \mid \mathbf{X}, \boldsymbol{\beta}^{(t)}, \sigma^2) \cdot p(\boldsymbol{\beta}^{(t)})}\right)$$

2. **Gibbs update for σ^2 :** Draw from the full conditional:

$$\sigma^2 \mid \text{rest} \sim \text{Inverse-Gamma}\left(a_0 + \frac{n}{2}, b_0 + \frac{1}{2} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2\right)$$

We ran 100,000 iterations with a burn-in of 10,000, initializing the sampler at the ordinary least squares (OLS) estimates. Posterior summaries, trace plots, and histograms were generated for all parameters.

3.4 INLA Implementation and Summary

The same model was fit using the **R-INLA** package. INLA is designed for latent Gaussian models (LGMs), where the likelihood depends on a latent Gaussian field $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$, and observations are conditionally independent given $\mathbf{x}$.

INLA approximates posterior marginals such as $p(\beta_j | \mathbf{y})$ and $p(\log \sigma^2 | \mathbf{y})$ using nested Laplace approximations. Unlike MCMC, INLA does not generate posterior samples; instead, it computes numerical approximations to each marginal posterior density.

For this project, we used INLA’s default priors for fixed effects and precision terms. Posterior means and 95% credible intervals were extracted from the `summary.fixed` object. Marginal densities were plotted using base R graphics, and predictions were made using the posterior means applied to the test set design matrix.

3.5 Evaluation Metrics

To assess model performance, we used the posterior means of the coefficients from each method to generate predictions on the test set. The following metrics were computed:

- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- Coefficient of determination (R^2)

Elapsed computation time for each method was measured using R’s `Sys.time()` function. Convergence of the MCMC sampler was assessed using trace plots and posterior histograms. INLA posterior densities were interpreted directly from the marginal approximations.

All code was written in R and is available in the appendix and GitHub repository for full reproducibility.

4 Conclusions

Both MCMC and INLA proved to be effective Bayesian inference methods for modeling the log-transformed housing price based on structural features of a home. The posterior summaries produced by each method were highly consistent in terms of sign, magnitude, and variable importance. Predictive performance on a held-out test set was nearly identical for both methods, demonstrating that either approach is suitable for modeling this type of continuous response.

The main distinction between the two methods was computational efficiency. While the MCMC implementation required over 90 seconds to generate 100,000 posterior samples, INLA completed its analysis in just over 13 seconds. This difference makes INLA particularly appealing for practical applications, especially when repeated model fits are needed or when working with larger datasets. INLA’s ability to provide smooth posterior marginals without sampling is also advantageous for rapid exploration and visualization.

One difficulty encountered during the project was tuning the MCMC proposal distribution to achieve an acceptable acceptance rate while ensuring sufficient exploration of the

parameter space. Trace plots and posterior histograms were necessary to verify convergence. INLA, by contrast, was much easier to apply and required no manual tuning. However, INLA is limited to models with latent Gaussian structure, and it offers less flexibility when extending to more custom model formulations.

For future work, several alternative modeling strategies could be explored:

- **Bayesian variable selection** to identify the most informative predictors and reduce model complexity.
- **Regularized regression methods**, such as ridge or Bayesian Lasso, to stabilize coefficient estimates in the presence of multicollinearity.
- **Nonlinear modeling techniques**, such as splines or polynomial terms, to capture curved relationships between predictors and price.
- **Posterior predictive checks** or model comparison criteria (e.g., WAIC) to assess model fit more thoroughly.

Overall, this project demonstrates the practical tradeoffs between exact and approximate Bayesian inference. MCMC provides full posterior sampling and greater model flexibility, while INLA offers a highly efficient and interpretable solution when the model assumptions are appropriate.

Table 5: Summary Comparison of MCMC and INLA

Aspect	MCMC	INLA
Posterior Accuracy	High	High
Credible Interval Width	Slightly wider	Narrower
Convergence Diagnostics	Trace plots required	Not applicable
Computation Time	91.43 sec	13.24 sec
Ease of Use	Manual tuning required	Fully automated
Flexibility	Very high	Moderate

5 Appendices

The dataset and all code used for this analysis are available at: https://github.com/lucretian-swerve/housing_mcmc_inla

All plots and summaries were generated using R.

6 Bibliography

Main Paper

Rue, H., Martino, S., & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(2), 319–392. <https://doi.org/10.1111/j.1467-9868.2008.00700.x>

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R-INLA Documentation

Rue, H., Martino, S., & Lindgren, F. (2023). The R-INLA project. Retrieved from <https://www.r-inla.org>